ALI GARABAGLU

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RESEARCH INTERESTS

Interested in advancing and applying statistical, machine learning, and physics-informed computational methods to problems across biology, physics, and data science.

EDUCATION

University of Washington, Seattle, WA **2026**
PhD in Physics
Thesis: "First Cross-Section Measurements of Highly Lorentz-Boosted Higgs Bosons in the di-tau Channel at ATLAS"
Advisor: Quentin Buat

Rutgers University, New Brunswick, NJ **2020**
B.S. magna cum laude in Physics with High Honors, Minor in Mathematics
Thesis: "A Study of the MATHUSLA Test Stand and Proposed Large Scale Geometries"
Advisor: John Paul Chou

SELECTED PAPERS

- ATLAS Collaboration, "Cross-section measurements of boosted Higgs boson production in final states with pairs of hadronically decaying τ-leptons with the ATLAS experiment". arXiv preprint arXiv:2607.22193 (2026). **(main contributor, thesis paper)**
- A. Garabaglu, Q. Buat and A. De Maria, "Identification of highly collimated hadronically decaying τ leptons", tech. rep., 2024, url: https://cds.cern.ch/record/2906623 **(main contributor)**
- FASER Collaboration, "First Measurement of ν_e and ν_μ Interaction Cross Sections at the LHC with FASER's Emulsion Detector", Physical Review Letters 133, 021802 (2024). https://doi.org/10.1103/PhysRevLett.133.021802 **(Core contributor)**
- FASER Collaboration, "First Measurement of the Muon Neutrino Interaction Cross Section and Flux as a Function of Energy at the LHC with FASER", Physical Review Letters 134, 211801 (2025). https://doi.org/10.1103/PhysRevLett.134.211801 **(Contributed to alignment of electronic detectors)**
- FASER collaboration, "Shining light on the dark sector: search for axion-like particles and other new physics in photonic final states with FASER". J. High Energ. Phys. 2025, 199 (2025). https://doi.org/10.1007/JHEP01(2025)199 **(Contributed to alignment of electronic detectors)**
- MATHUSLA collaboration, "Recent Progress and Next Steps for the MATHUSLA LLP Detector". arXiv preprint arXiv:2203.08126 (2022). **(Core contributor)**
- MATHUSLA collaboration, "The MATHUSLA test stand." Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment 985 (2021): 164661 **(Core contributor)**
- MATHUSLA collaboration, "An Update to the Letter of Intent for MATHUSLA: Search for Long-Lived Particles at the HL-LHC." arXiv preprint arXiv:2009.01693 (2020). **(main contributor)**

RESEARCH EXPERIENCE

- **Postdoctoral Associate, Yale University** **September 2026-present**

Developing physics-informed generative models and optimal transport methods to study cellular dynamics, from foundational theory to algorithms and software.

- **Visiting Research Assistant, Yale University** **July 26'-August 26'**

Developing machine learning and optimal transport methods to identify structural similarity in protein micro-environments, with applications to kinase classification, metal-binding site identification, and other problems in biology.

- **Research Assistant, University of Washington** **2021-2026**

Worked on data extraction and analysis from the FASER emulsion detector and theoretical modeling of signal events during the first two years, before transitioning to the ATLAS experiment where I focused on developing ML based identification and reconstruction algorithms and leading the boosted Higgs to di-tau analysis.

- **Research Assistant, Rutgers University** **2017-2021**

Developed and implemented Monte Carlo simulations, and software frameworks for physics analyses

TEACHING

- Teaching assistant for High Energy Physics — Winter 2023
- Teaching assistant for Neural Network Methods for Signals in Engineering and Physical Sciences — Spring 2022(23)
- Teaching assistant for Digital Electronics Lab — Spring 2022
- Teaching assistant for Mathematical Physics — Winter 2022
- Teaching assistant for Computational Data Analysis Lab — Autumn 2022
- Teaching assistant for Nuclear & Particle Physics Experiments Lab — Autumn 2021

TALKS

- *First ATLAS Measurements of Highly Lorentz-Boosted Higgs Bosons Decaying to Taus*, 2026, APS Division of Particles and Fields Meeting, Fermilab, IL
- *Omni: Identification of Highly Collimated DiTau Objects*, 2026, Tau Combined Performance group workshop, Geneva, Switzerland
- *Measurement of the fully hadronic $H \rightarrow \tau\tau$ cross-section with the ATLAS experiment*, 2026, LHCP, Paris, France
- *Boosted $H \rightarrow bb/cc/\tau\tau$ tagging*, 2025, ATLAS Hadronic Identification Workshop, Geneva, Switzerland
- *Reconstruction and Identification of Highly Collimated DiTau Objects*, 2025, Tau Combined Performance group workshop, Geneva, Switzerland
- *Probing the High-Momentum Frontier of Higgs Boson Decay to a pair of Tau Leptons*, 2024, ATLAS US Workshop, Seattle, WA
- *Boosted DiTau High + Low pt Reconstruction and Identification*, 2024, TauCP + HLepton Workshop, Geneva, Switzerland
- *Results From TeV Neutrinos at the FASER Experiment*, 2024, DPF-PHENO, Pittsburg, PA
- *FASER Tracking and Emulsion Station Alignment*, 2024, ACAT Conference, Stony Brook, NY
- *Weakly-Supervised Anomaly Detection with Conditional VAEs*, 2023, APS, online

AWARDS

- University of Washington Provost Award — 2021
- Rutgers University SAS Paul Robeson Scholar — 2020
- Rutgers University Robert L. Sells Scholarship — 2020
- Rutgers University SAS Excellence Award — 2019
- Rutgers University Jaqua Scholarship — 2019